VANSH TOMAR

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ABOUT ME

Mathematics and Computing undergraduate with hands-on project experience in machine learning, scientific data processing, and LLM-powered retrieval systems. Interested in developing practical AI solutions, with a particular focus on agentic AI, deep learning, and geospatial applications.

EDUCATION

Madhav Institute of Technology and Science — Gwalior, India
B.Tech. in Mathematics and Computing (CGPA: 9.17) — *Aug 2025 – Expected Aug 2029*

Swami Vivekananda Academy — Gwalior, India
CBSE Class 12th PCM, Percentage: ***91.8%*** — *Apr 2023 – Jun 2024*

PROJECTS

Agentic Scientific Research Assistant — *Python, RAG, ChromaDB, Sentence Transformers, Groq API*

- Developed an agentic RAG system for scientific research, integrating retrieval from oceanographic research papers with structured oceanographic data.
- Implemented document retrieval using BM25 and vector search, followed by cross-encoder reranking to improve the relevance of retrieved research passages.
- Built an LLM-based query router using Groq API to direct queries to paper retrieval, dataset analysis, or combined workflows, enabling evidence-based responses.

Neural Network from Scratch — *Python, NumPy*

- Implemented a neural network from scratch in Python and NumPy to understand the mathematical foundations of forward propagation and backpropagation.
- Implemented core components including weighted layers, activation functions, loss computation, gradient calculation, and parameter updates.
- Trained and evaluated the model while analyzing learning behavior and model performance without relying on high-level deep learning frameworks.

Urban Heat Mitigation using AI/ML — *Python, Google Earth Engine, Remote Sensing*

- Developing an ML-based approach to analyze urban heat patterns in Delhi using satellite-derived environmental and urban features.
- Processed Landsat and Sentinel-2 imagery to derive Land Surface Temperature (LST), NDVI, NDWI, EVI, and NDBI.
- Engineered additional geospatial features including land cover, tree cover, population density, elevation, slope, and distance to water for urban heat analysis and cooling strategy evaluation.

SKILLS

Programming: Python, C++

AI/ML: Machine Learning, Neural Networks, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI

Libraries & Frameworks: NumPy, Pandas, Scikit-learn, Sentence Transformers, ChromaDB, FastAPI, Langchain

Tools: Git, GitHub, Jupyter Notebook, VS Code